Valentin Py

Aerospace Engineering Student - Computational Fluid Dynamics & Applied Mathematics

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Summary

Aerospace engineering student at ISAE-SUPAERO within the Propulsion and Combustion track, I made an exchange semester focused on numerical methods for fluid dynamics at the University of Tokyo and followed the M2RI at Université Paul Sabatier.

During my previous internships I developped skills about programming (C++/Java/Python), numerical methods for fluid dynamics (Real-time methods SPH/FLIP-PIC or Riemann solvers) and put hands on simulating software for diffferent CFD cases (STARCCM+/OpenFOAM).

My goal is to strenghened my knowledge at the intersection between applied mathematics and numerical methods for CFD. Driven by rigor and a desire to build next-generation simulation tools, I welcome connections with researchers and laboratories in these fields.

Experience

ISAE-SUPAERO - Department of Aeronautics (DAEP), Research & Development Intern - Scientific Visualization & AR/VR — Toulouse, France, Apr 2026 – present

- Conducted aerodynamic simulations (RANS/URANS) on a NACA0012 airfoil using STAR-CCM+, focusing on flow physics analysis for teaching key concepts of aerodynamics.
- Engineered a real-time VR/AR visualization framework within Unreal Engine 5 using Blueprints & post-processing softwares like Paraview and Blender to render aerodynamic criteria (e.g., Q-criterion, velocity fields, streamlines).
- Developped and explored different numerical methods in C++ for real-time fluid dynamics simulations around NACA0012 airfoil (LBM method through GPU computing).

Ascendance, Software Engineer Intern on ICD — Toulouse, France, Mar 2025 – Aug 2025

- Solo developed, tested, and integrated a software using Python and JavaFX to manage the Interface Control Documents (ICD) for a hybrid-electric aircraft.
- Designed automated validation and verification (V&V) workflows to bridge the gap between ICD specifications and hardware test benches using custom Python automation scripts.
- Contributed to an innovative thermal-electric propulsion project, strengthening skills in analyzing multidisciplinary data coupling and complex systems engineering.

Hinfact, Internship - Data Management & Process Digitization — Toulouse, France, June 2024 – Aug 2024

- Structured database and web interfaces to centralize and process training data for airline pilots.
- Architected information management tools (inventories, functional guides, relational databases) using Notion.
- Developed competencies in data curation, information architecture, and process digitization within a fast-paced aerospace tech environment.

Skills

CFD & Numerical Methods: FDM, FVM, LBM, Galerkin Method, Turbulence Modeling, Mesh Convergence, Scalar Conservation Laws

CFD Software: STAR-CCM+, OpenFOAM, Ansys

HPC & GPU Computing: MPI, OpenMP, CUDA, OpenCL

Scientific Programming: Python (PyVista, NumPy), C++, Java

Scientific Visualization: ParaView, Blender, Unreal Engine 5, OpenGL, AR/VR pipelines

Tools: Git, Cycle en V, Notion

Education

Université Toulouse III - Paul Sabatier, MSc in Master 2 Recherche et Innovation (M2RI) - Parcours Déterministe, EDP et Simulations Numériques — Toulouse, France, Sept 2026 – Aug 2027

- Coursework: An Introduction to the Theoretical and Numerical Analysis of Scalar Conservation Laws; Controlled Dynamical Systems: Structured Modelling and Numerical Methods; Regularization of Ill-posed Inverse Problems and Applications; High-Performance Computing (MPI, OpenMP, CUDA).
- Double-degree track with the 3rd-year ISAE-SUPAERO curriculum, focused on rigorous numerical analysis of PDEs applied to CFD.

University of Tokyo, Aerospace, Aeronautical and Astronautical Engineering (Exchange Semester) — Tokyo, Japan, Sept 2025 – Mar 2026

- Coursework: Advanced Theoretical Numerical Simulation; Discretization Schemes (FDM/FVM); Lattice Boltzmann Method (LBM); Fluid-Structure Interaction (FSI) applied to bioengineering frameworks.
- Academic project: high-fidelity aerodynamic simulations on a NACA 0012 airfoil using OpenFOAM, analyzing mesh convergence and solver stability.

ISAE-SUPAERO, MEng in Aerospace, Aeronautical and Astronautical Engineering - Fluid Dynamics / Internal Aerodynamics (Propulsion, Combustion) & Applied Mathematics — Toulouse, France, 2023 – 2027

- Core coursework: advanced multi-physics fluid dynamics (unsteady, 3D, aeroacoustics, aeroelasticity), numerical simulation and turbulence (instabilities, turbulence modeling, numerical methods).
- Turbomachines & Combustion track: turbomachinery aerodynamics, two-phase flows and combustion, design case studies.
- Technical expertise: hands-on experience with industry-standard CFD software and solvers (Ansys, STAR-CCM+, OpenFOAM) for complex flow analysis and turbulence modeling.

Extracurricular Activities

OSE ISAE-SUPAERO Program

Toulouse, France | September 2023 - February 2025

- Communications & Outreach Officer: member of the student team leading the OSE (Ouverture Sociale Étudiante) program, communicating the program's actions and promoting equal opportunities among youth (covered events, wrote articles detailing the latest initiatives).
- Student Tutor: tutoring and academic mentoring for high-school students (Seconde to Terminale) as part of a "cordées de la réussite" outreach program.

SCUBE Rocketry Club - ISAE-SUPAERO

Toulouse, France | September 2023 - February 2025

- Treasurer of the school's rocketry club.
- Completed a 5-month training program leading to the launch of a mini-rocket with CNES (aerodynamics, propulsion, Python programming).

Arts & Travel Association - ISAE-SUPAERO
Toulouse, France | September 2023 - February 2025
- President of the travel branch, in charge of organizing cultural trips abroad.
- Organized cultural trips in several European countries (Italy, Spain).

Projects

SPH Fluid Solver (C++/OpenGL)
Weakly-compressible smoothed-particle hydrodynamics solver (cubic spline kernel, Tait equation of state), validated on the classic dam-break benchmark and rendered in real time with a custom OpenGL engine.
- valentinpy5-spec.github.io/projects/sph_dam_break

Exact Riemann Solver for the 1D Euler Equations
Finite volume solver for the compressible Euler equations with an exact Riemann solver (Newton-Raphson star-region pressure) and RK2 time integration, validated on the Sod shock tube.
- valentinpy5-spec.github.io/projects/1_project

NACA0012 Flow Solver (LBM, GPU)
Real-time Lattice-Boltzmann (D2Q9/BGK) solver in OpenCL for 2D flow around a NACA0012 airfoil, with live angle-of-attack control and OpenGL visualization.
- valentinpy5-spec.github.io/projects/naca0012_lbm_gpu

Interests

Hobbies
Trail running, Climbing, Swimming

Languages

French: Native speaker

English: Fluent

Japanese: Beginner

German: Beginner